Estimating D_x from Noisy Droplet Counts

Sampling noise, the Poisson loss, and error bars
for the steady-state problem

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1 From exact data to counted droplets

In the noise-free experiment, the data were the model's exact size distribution, and the method recovered the true diffusivity D_x to about nine digits. Real measurements are not like that. An

instrument looks at a small volume of chamber air and *counts* the droplets it finds there, sorted into size bins. This note adds that one ingredient, **sampling noise**, and works out what changes.

Why counting creates noise. We assume the chamber is *well mixed*: on average, every part of it contains the same size distribution. So a sample taken anywhere is representative *on average*. But a small sample contains only a limited number of droplets, and exactly how many of each size happen to be inside it at that moment is a matter of chance. A poll is a good analogy: if people are thoroughly mixed, a random poll is unbiased, but a poll of 20 people still gives noticeably different answers each time. Well-mixed removes *bias*; it does not remove this luck-of-the-draw *noise*.

When it matters. The noise is large when samples are small or bins are sparsely populated, as with instrument data, especially for the rare large droplets. It is negligible when a sample contains an enormous number of droplets, as when a simulation reports every droplet in its domain.

What changes. Very little of the machinery: the model, the adjoint, and the optimizer are the same as before. The loss changes, and so does one vector in the adjoint calculation. The bigger change is in the *question*. With noisy data we cannot recover D_x exactly, so the goal becomes an estimate *with an honest error bar*. Sections 3–7 develop the theory; Section 8 tests it.

2 The model

We describe droplet sizes by $x = r^2$ (μm^2). The size distribution $N(x)$ ($\text{cm}^{-3}\,\mu\text{m}^{-2}$) is defined so that $N(x)\,\mathrm{d}x$ is the number of droplets per cm^3 of air with r^2 between x and $x + \mathrm{d}x$. In a steady, well-mixed chamber it satisfies

$$\frac{\mathrm{d}F}{\mathrm{d}x} + \lambda(x)\,N = 0, \qquad F = A(x)\,N - D_x\,\frac{\mathrm{d}N}{\mathrm{d}x}, \qquad F(x_{\min}) = F_{\text{in}}, \quad F(x_{\max}) = 0. \tag{1}$$

Here F is the flux of droplets through size x; $A(x) = a\,(s - s_{\text{eq}}(r))$ is the condensational growth rate, with Köhler equilibrium supersaturation $s_{\text{eq}} = A_k/r - B_k/r^3$; D_x is the turbulent broadening we want to estimate; $\lambda(x) = kx$ is the rate at which droplets fall out; and $F_{\text{in}} = 100\ \text{cm}^{-3}\text{s}^{-1}$ is the rate at which new droplets are injected at the smallest size.

We solve (1) with finite volumes: $n = 8000$ equal cells of width $\Delta x = 0.1125\,\mu\text{m}^2$, spanning r from 0.0325 to $30\,\mu$m, with Scharfetter–Gummel fluxes between cells. The result is a tridiagonal linear system

$$\mathsf{A}(D_x)\,\mathbf{N} = \mathbf{b}, \qquad \mathbf{b} = (F_{\text{in}}, 0, \ldots, 0)^{\top}, \tag{2}$$

where N_i is the average of $N(x)$ over cell i, so $N_i\,\Delta x$ is the concentration of droplets in that cell. Parameter values are in Appendix A.

3 What the instrument sees

3.1 Sample volume, bins, and expected counts

The instrument examines a volume V (cm^3) of chamber air and counts the droplets in each size bin $[x_k, x_{k+1}]$, $k = 1, \ldots, K$. The data are whole numbers,

$$c_k = \text{number of droplets counted in bin } k.$$

For simplicity we make each bin exactly m consecutive model cells, so bin edges always fall on cell edges (Figure 1). We use $m = 8$, giving $K = 1000$ bins of width $0.9\,\mu\text{m}^2$.

Because the chamber is well mixed, the expected number of droplets in the sample is the sample volume times the concentration. For bin k,

$$\mu_k = V \int_{x_k}^{x_{k+1}} N(x)\,\mathrm{d}x = V\,\Delta x \sum_{i\,\in\,\text{bin } k} N_i, \qquad \text{or} \qquad \boldsymbol{\mu} = V\,\mathsf{H}\,\mathbf{N}. \tag{3}$$

Units: $\text{cm}^3 \times \text{cm}^{-3}\mu\text{m}^{-2} \times \mu\text{m}^2$ is a pure number, a count. The matrix H adds $N_i\,\Delta x$ over the cells of each bin; if each bin were a single cell ($m = 1$), the sum would have one term and $\mathsf{H} = \Delta x\,\mathsf{I}$. Note that μ_k is an *average*: an actual sample contains more or fewer droplets than μ_k by chance.

The sample volume sets how many droplets are counted in total, $V \sum_i N_i\,\Delta x$. With the reference concentration of about 20,600 cm^{-3} (for $D_x = 10$), samples of $V = 0.01$, 0.1 and 1 cm^3 hold about 200, 2000 and 20,000 droplets. Figure 2 shows what such samples look like.

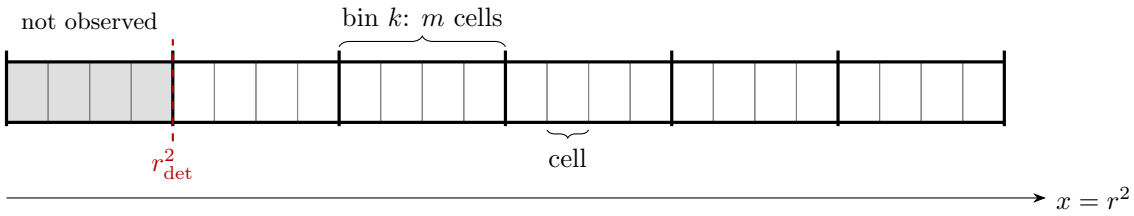


Figure 1: Model cells (thin lines) grouped into observation bins of m cells (thick lines). Cells below the detection limit (shaded) belong to no observed bin.

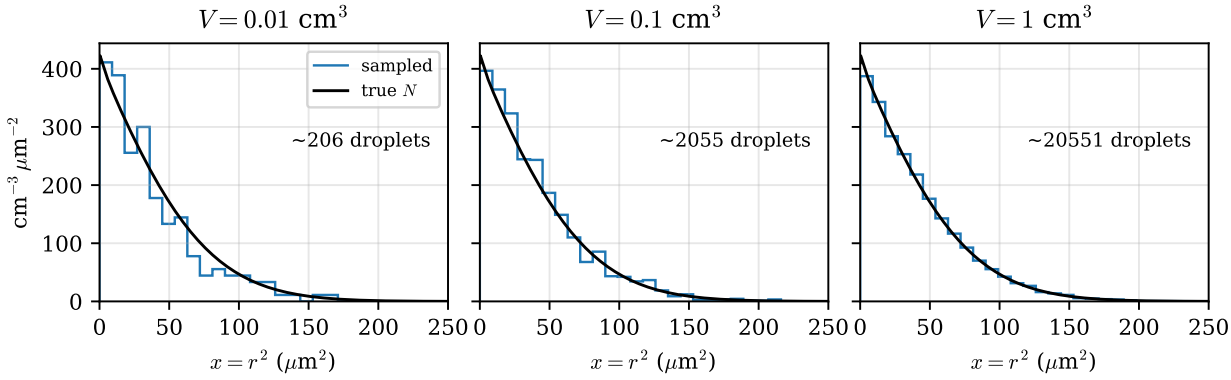


Figure 2: One random sample at each of three sample volumes (true $D_x = 10$), converted back to concentration (counts divided by V and bin width) and compared with the true N. For display only, these use wider bins of 80 cells. Small samples are visibly noisy, especially in the tail.

3.2 The detection limit: unobserved is not the same as zero

Real instruments cannot see droplets smaller than some radius r_{det}. It's tempting to treat those bins as "observed, zero droplets." That would be wrong. There *are* plenty of droplets there; we simply did not look. Recording zeros would tell the fit "we looked and found nothing," and the fit would distort D_x to suppress small droplets.

The correct treatment is to **leave those bins out of the data entirely**. The model still covers the full size range, because small droplets exist and feed the larger ones. Only the bins lying entirely above r_{det} are compared with data (Figure 1).

4 How the counts are distributed

4.1 One bin: the Poisson distribution

Suppose the chamber holds M droplets in bin k. Because the chamber is well mixed, each of them is inside the sampled volume with the same small probability $q = V/V_{\text{chamber}}$, independently of the others. Counting how many land in the sample is like flipping M coins that each come up heads with probability q. The count is therefore *binomial*:

$$P(c) = \binom{M}{c} q^c \, (1-q)^{M-c}, \qquad \text{with mean } Mq = \mu_k.$$

In a chamber, M is enormous and q is tiny (cubic centimeters out of cubic meters). Let $M \to \infty$ and $q \to 0$ with $Mq = \mu$ held fixed, and look at the pieces one at a time:

$$\binom{M}{c} q^c (1-q)^{M-c} = \underbrace{\frac{M(M-1)\cdots(M-c+1)}{M^c}}_{\to 1} \cdot \frac{\mu^c}{c!} \cdot \underbrace{\left(1 - \frac{\mu}{M}\right)^{M}}_{\to e^{-\mu}} \cdot \underbrace{\left(1 - \frac{\mu}{M}\right)^{-c}}_{\to 1}.$$

The limit is the **Poisson distribution**:

$$P(c_k \mid D_x) = \frac{\mu_k^{c_k} \, e^{-\mu_k}}{c_k!}, \qquad c_k = 0, 1, 2, \ldots \tag{4}$$

The model enters only through the expected count μ_k. The Poisson distribution has a key property: **its variance equals its mean**. A bin expecting μ droplets typically misses by about $\sqrt{\mu}$, so the *relative* error is $1/\sqrt{\mu}$ (Figure 3). A bin expecting 10,000 droplets is accurate to about 1%; a bin expecting 4, only to about 50%.

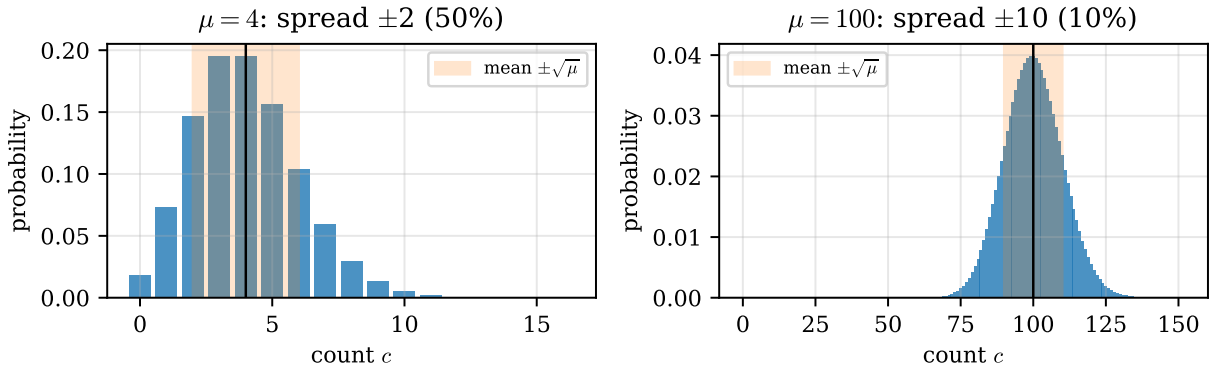


Figure 3: Poisson distributions with expected counts 4 and 100. The typical deviation is $\sqrt{\mu}$, so relative noise shrinks as counts grow.

4.2 Many bins: independence

Different bins count different droplets, and in the same limit their counts are *independent*. Formally: if the total count is Poisson and each droplet independently falls in bin k with some probability, the bin counts are independent Poisson variables.

It's worth separating two ideas that are easy to confuse:

- **Neighboring bins have similar counts.** True, but that is about the *expected* counts: the true distribution is smooth, so neighboring μ_k are similar. The model already accounts for this.
- **Neighboring bins have correlated *errors*.** This is what independence rules out. If two bins each expect 100 droplets and one happens to get 115, under the Poisson model that says nothing about whether the other gets more or less than 100.

Correlated errors do appear in real data when something *shared* shifts many bins at once: the sample happens to come from a pocket of air with more droplets, the chamber's supersaturation is momentarily high so the whole distribution shifts, or the sample volume is slightly wrong. These make real counts more variable than Poisson predicts. They are physics and instrument effects beyond pure sampling, and we leave them for later. In the synthetic experiments here, independence holds exactly, because we draw each bin's count independently.

5 The loss function

5.1 Derivation

Likelihood. By independence, the probability of the whole observed histogram is a product over the observed bins:

$$L(D_x) = \prod_{k \in \text{obs}} \frac{\mu_k^{c_k}\, e^{-\mu_k}}{c_k!}.$$

Read as a function of D_x with the data fixed, this is the **likelihood**: how probable the observed counts would be if the true diffusivity were D_x. The natural estimate is the value that makes the data most probable, the **maximum-likelihood estimate**.

Negative log-likelihood. Maximizing L is the same as minimizing $-\ln L$ (the logarithm is increasing), which turns the product into a sum:

$$-\ln L = \sum_{k \in \text{obs}} \left[\mu_k - c_k \ln \mu_k + \ln c_k!\right].$$

The last term depends only on the data, not on D_x, so we can drop it.

Subtract the best possible fit. If each μ_k could be chosen freely, a term would be smallest where $\frac{\mathrm{d}}{\mathrm{d}\mu}(\mu - c\ln\mu) = 1 - c/\mu = 0$, i.e. at $\mu = c$. Subtracting this best-possible value and multiplying by 2 gives the **Poisson deviance**, which we use as the loss:

$$\boxed{\mathcal{J}(\ell) = 2 \sum_{k \in \text{obs}} \left[c_k \ln \frac{c_k}{\mu_k(\ell)} - c_k + \mu_k(\ell)\right], \qquad 0 \ln 0 \equiv 0, \qquad \ell = \ln D_x.} \tag{5}$$

The subtracted piece doesn't depend on D_x, so $\mathcal{J}$ has the same minimizer as $-\ln L$. The factor 2 makes the curvature formula of Section 7 simple. As before, we estimate $\ell = \ln D_x$, so D_x stays positive and changes are measured as ratios.

5.2 Properties

- **Never negative; zero only for a perfect fit.** For fixed $c \geq 0$, the term $f(\mu) = c\ln(c/\mu) - c + \mu$ has $f'(\mu) = 1 - c/\mu$ and $f''(\mu) = c/\mu^2 \geq 0$. So it is convex, with its minimum $f(c) = 0$ at $\mu = c$. Hence $\mathcal{J} \geq 0$, with equality only when every $\mu_k = c_k$. With noisy data, even the true D_x cannot match random counts exactly, so the loss at the minimum is positive.
- **Empty bins are handled correctly.** A bin with $c_k = 0$ contributes $2\mu_k$, penalizing the model for predicting droplets that weren't seen.
- **Relation to least squares.** When counts are large, write $c = \mu + \varepsilon$ and expand $\ln(1 + \varepsilon/\mu)$ to second order:

$$2\Big[(\mu+\varepsilon)\ln\Big(1+\frac{\varepsilon}{\mu}\Big) - \varepsilon\Big] \approx \frac{(c-\mu)^2}{\mu}.$$

  So the deviance is *weighted* least squares, with each bin weighted by $1/\mu_k$, the inverse of its noise variance. Plain least squares, used in the noise-free experiment, treats every bin as equally reliable. It lets the high-count bins near the peak dominate and nearly ignores the sparse tail bins, where much of the broadening signal lives. The exact form (5) remains correct for bins with only a few counts, where this approximation fails.

6 The gradient and the optimizer

The adjoint method needs only $\mathbf{g} = \partial\mathcal{J}/\partial\mathbf{N}$. Differentiating (5) and using $\boldsymbol{\mu} = V\mathsf{H}\mathbf{N}$:

$$\frac{\partial\mathcal{J}}{\partial\mu_k} = 2\Big(1 - \frac{c_k}{\mu_k}\Big) \quad\Longrightarrow\quad \mathbf{g} = V\,\mathsf{H}^\top\Big[2\big(1 - c_k/\mu_k\big)\Big]_k, \tag{6}$$

with zero for unobserved bins. In words: each cell receives its bin's residual weight $2(1 - c_k/\mu_k)$, times $V\Delta x$. The weight is zero where the model matches the count, positive where it predicts too many droplets, and negative where it predicts too few. Everything else is unchanged from the noise-free case:

$$\mathsf{A}^\top\boldsymbol{\psi} = -\mathbf{g}, \qquad \frac{d\mathcal{J}}{dD_x} = \boldsymbol{\psi}^\top\frac{d\mathsf{A}}{dD_x}\mathbf{N}, \qquad \frac{d\mathcal{J}}{d\ell} = D_x\,\frac{d\mathcal{J}}{dD_x}. \tag{7}$$

Cells below the detection limit get $g_i = 0$, but they still take part in the adjoint solve, so the gradient correctly accounts for how they feed the observed sizes.

Loss and gradient for noisy counts

1. $D_x = e^\ell$. Solve $\mathsf{A}\mathbf{N} = \mathbf{b}$.
2. Expected counts $\mu_k = V\Delta x\sum_{i\in k} N_i$; loss $\mathcal{J}$ from (5) over observed bins.
3. $\mathbf{g}$ from (6); adjoint solve $\mathsf{A}^\top\boldsymbol{\psi} = -\mathbf{g}$, reusing the factorization of A.
4. Return $\mathcal{J}$ and $d\mathcal{J}/d\ell = D_x\,\boldsymbol{\psi}^\top(d\mathsf{A}/dD_x)\mathbf{N}$.

Optimizer. We use the same gradient descent as in the noise-free case: a step size estimated from the last two gradients, a cap of one unit in ℓ per iteration, and backtracking until the loss decreases. Three small changes:

- **Bounds.** D_x is restricted to $[10^{-3}, 10^3]\,\mu\mathrm{m}^4/\mathrm{s}$. With very little data, the best fit can wander far off.

- **A looser stopping rule.** Stop when $|d\mathcal{J}/d\ell| < 10^{-7}$. There is no point chasing digits far beyond the statistical precision.
- **No normalization.** The loss is not divided by anything.

Every fit starts from $D_x = 1$.

7 Error bars from a single sample

With noise, each new sample gives a slightly different estimate $\hat{\ell}$. With synthetic data we can simply repeat the experiment many times and measure the scatter directly (Section 8.2). With real data we usually have one sample, so we need an error bar from that sample alone.

The idea. Near its minimum, the loss is approximately a parabola in ℓ:

$$\mathcal{J}(\ell) \approx \mathcal{J}(\hat{\ell}) + \tfrac{1}{2}\,\mathcal{J}''(\hat{\ell})\,(\ell - \hat{\ell})^2.$$

Since $\mathcal{J} = 2(-\ln L) + \text{const}$, the likelihood is then approximately

$$L(\ell) \propto e^{-\mathcal{J}(\ell)/2} \propto \exp\Big(-\frac{(\ell - \hat{\ell})^2}{2\,\sigma^2}\Big), \qquad \sigma^2 = \frac{2}{\mathcal{J}''(\hat{\ell})}.$$

This is a Gaussian (bell curve) in ℓ centered at the estimate, with standard deviation σ. A sharply curved loss means the data pin ℓ down tightly; a flat loss means they don't. So the error bar is

$$\boxed{\operatorname{sd}(\hat{\ell}) \approx \sqrt{\frac{2}{\mathcal{J}''(\hat{\ell})}}} \tag{8}$$

We compute $\mathcal{J}''$ by finite-differencing the adjoint gradient: $\mathcal{J}'' \approx [g(\hat{\ell} + h) - g(\hat{\ell} - h)]/(2h)$ with $h = 10^{-3}$, which costs two extra gradient evaluations. Equivalently, ± 1 sd is where the loss rises by 1 above its minimum (Figure 6). Because $\ell = \ln D_x$, the result reads as a *relative* error: $\operatorname{sd}(\hat{\ell}) = 0.06$ means D_x is known to about 6%, i.e. to within a factor $e^{\pm 0.06}$.

The predicted error, and where the information comes from. Averaging the curvature over many possible samples gives the **Fisher information**,

$$\mathcal{I} = \sum_{k \,\in\, \text{obs}} \frac{1}{\mu_k}\Big(\frac{\partial \mu_k}{\partial \ell}\Big)^2, \qquad \operatorname{sd}(\hat{\ell}) \approx \frac{1}{\sqrt{\mathcal{I}}}. \tag{9}$$

This shows where information comes from: each bin contributes its sensitivity to D_x, squared, divided by its own noise level μ_k. It also shows how precision scales with the sample. Every μ_k is proportional to V, so $\mathcal{I} \propto V$ and

$$\operatorname{sd}(\hat{\ell}) \propto \frac{1}{\sqrt{V}}\,:$$

four times the sample volume halves the error bar. This is the data-sparsity question in quantitative form. We use (9), evaluated at the true D_x, as the *predicted* spread in the experiments.

bin width (μm^2)	0.11	0.9	9	90
$D_x = 10$, $V = 0.1$ cm^3	6.1%	6.1%	6.1%	6.7%
$D_x = 0.1$, $V = 1$ cm^3	5.9%	7.1%	8.4%	13.2%

Table 1: Predicted relative error $1/\sqrt{\mathcal{I}}$ for different bin widths, with no detection limit.

Bin width. Merging bins can only lose information, since it discards where within the bin each droplet was. Table 1 shows how much, using (9). For $D_x = 10$, bins up to 9 μm^2 wide lose essentially nothing. For $D_x = 0.1$, whose fingerprint is fine structure near 1–2 μm, even the 0.9 μm^2 bins used here cost about 20% in precision. The Poisson loss is correct even for very fine, mostly empty bins, so finer bins never hurt.

8 Experiments

Generating noisy data. Solve (2) at the true D_x, compute $\boldsymbol{\mu}^{\text{true}} = V\mathsf{H}\mathbf{N}^{\text{true}}$, and draw each count independently from a Poisson distribution with that mean. The truth and the fit use the same grid; the only difference between data and model is sampling noise.

8.1 Sanity check: the new loss on exact data

Before adding noise, we replaced least squares by the deviance and used the exact expected counts as data. From starting guesses spanning 10^{-3} to 10^{3}, the fit recovered $D_x = 10$ to a relative error of about 10^{-11}, with $\mathcal{J} \approx 10^{-14}$ at the solution. This isolates the change of loss from the change of data.

8.2 Many samples: bias and spread

For each combination of true $D_x \in \{10, 0.1\}$ and $V \in \{0.01, 0.1, 1\}$ cm^3, we drew 200 independent samples, fit each one, and recorded $\hat{\ell}$. Two questions:

- **Bias:** is the average estimate on the truth?
- **Spread:** does the scatter of $\hat{\ell}$ match the prediction $1/\sqrt{\mathcal{I}}$?

Table 2 and Figures 4–5 show the results.

true D_x	V (cm^3)	droplets	predicted spread	observed spread	bias	single-sample sd (mean)
10	0.01	207	0.192	0.179	-0.017 ± 0.013	0.192
10	0.1	2053	0.061	0.064	0.000 ± 0.004	0.061
10	1	20558	0.019	0.018	-0.002 ± 0.001	0.019
0.1	0.01	344	0.712	0.990	-0.147 ± 0.070	0.758
0.1	0.1	3443	0.225	0.241	-0.029 ± 0.017	0.228
0.1	1	34387	0.071	0.079	$+0.006 \pm 0.006$	0.071

Table 2: 200 samples per row. "Spread" is the standard deviation of $\hat{\ell} = \ln \hat{D}_x$ (approximately the relative error of $\hat{D}_x$); "bias" is the mean of $\hat{\ell} - \ell_{\text{true}}$ with its standard error; the last column averages the curvature error bar (8) over the 200 fits. With 200 samples, an observed spread is itself uncertain by about 5%.

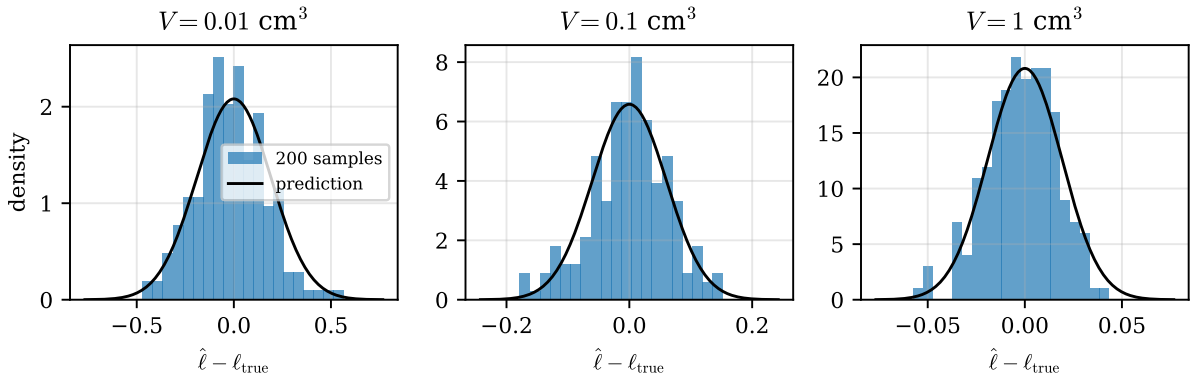


Figure 4: Distribution of $\hat{\ell} - \ell_{\text{true}}$ over 200 samples (true $D_x = 10$), with the predicted Gaussian of standard deviation $1/\sqrt{\mathcal{I}}$. Note the shrinking horizontal scale as V grows.

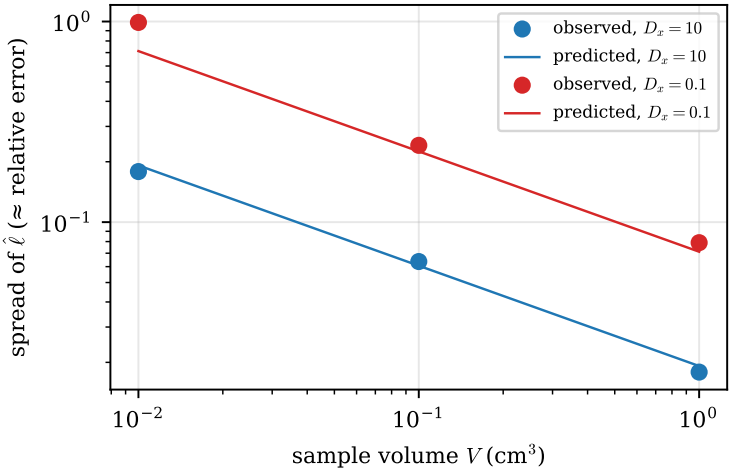


Figure 5: Observed spread (dots) and predicted spread (lines) versus sample volume. The lines have slope $-1/2$: ten times more sample gives about three times less error.

What we see.

- **For $D_x = 10$, theory and experiment agree throughout.** The estimates are unbiased, and the observed spread matches the prediction within the statistical uncertainty of 200 samples. The relative error is about 19%, 6% and 2% for roughly 200, 2000 and 20,000 droplets.
- **$D_x = 0.1$ is harder.** For the same number of droplets, the error is about five times larger, because weak broadening changes the distribution less. With about 3400 or 34,000 droplets, the observed spread is within about 10% of the prediction.
- **The prediction fails when the information is very low.** At $D_x = 0.1$ with only about 340 droplets, the predicted error is 71%. The observed spread is 99%, the estimates are skewed toward small values, and 4 of 200 fits ran to the lower bound. The parabola picture of Section 7 assumes the loss is close to a parabola over the range of plausible values, and when the error is comparable to the value itself, it isn't. In that regime, the error bar from

(8) should be treated as optimistic.

- **The $1/\sqrt{V}$ law holds** (Figure 5).

8.3 One sample: does its error bar tell the truth?

Figure 6 shows a single sample with $V = 0.1$ cm^3 and true $D_x = 10$. The fit gives $\hat{D}_x = 9.88$ after 9 iterations. The curvature error bar is $\mathrm{sd}(\hat{\ell}) = 0.061$, so the estimate is 9.88 within a factor $e^{\pm 0.061}$, i.e. [9.30, 10.50], which contains the truth. The deviance is indeed close to a parabola over this range.

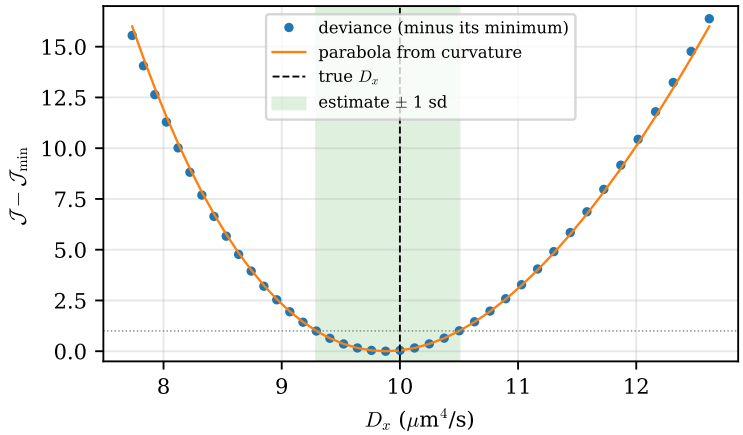


Figure 6: One sample: the deviance near its minimum (dots), the parabola implied by its curvature (line), and the resulting ±1 sd interval (shaded), where the loss rises by 1 (dotted line).

One sample's error bar can't be judged in isolation. The meaningful test is: over many samples, does the interval $\hat{\ell} \pm \mathrm{sd}$ contain the truth about 68% of the time, and $\hat{\ell} \pm 2\,\mathrm{sd}$ about 95% of the time, as a Gaussian error bar should? Figure 7 shows the first 40 samples: 27 of 40 intervals (68%) contain the truth. Table 3 gives the full results.

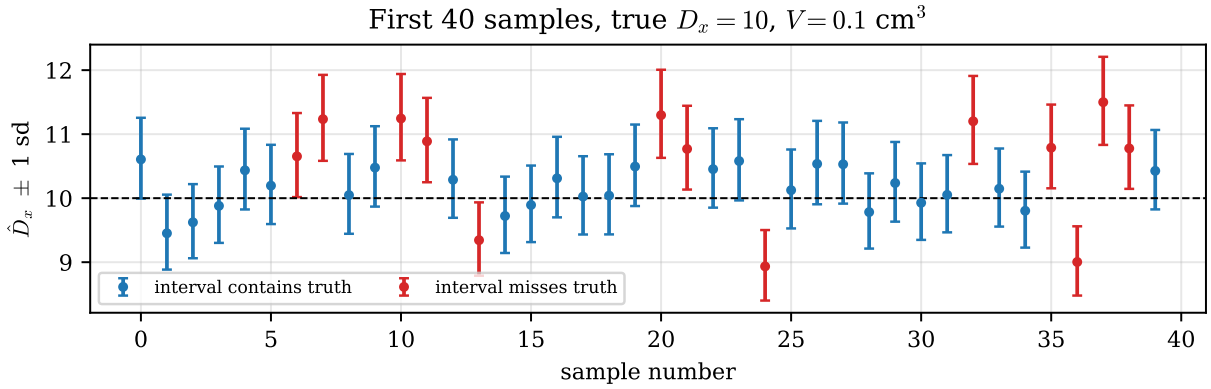


Figure 7: Estimates and their single-sample ±1 sd error bars for the first 40 samples. About two-thirds of the intervals should contain the truth (blue), and about one-third should miss it (red).

true D_x	V (cm^3)	observed spread	mean single-sample sd	within ±1 sd	within ±2 sd
10	0.01	0.179	0.192	71%	97%
10	0.1	0.064	0.061	69%	93%
10	1	0.018	0.019	72%	97%
0.1	0.01	0.990	0.758	67%	95%
0.1	0.1	0.241	0.228	62%	94%
0.1	1	0.079	0.071	64%	91%
			ideal (Gaussian)	68%	95%

Table 3: Do single-sample error bars tell the truth? Fraction of 200 samples whose interval contains the true value.

The single-sample error bars are accurate: on average they match the scatter actually seen across samples, and they contain the truth about as often as they should. For $D_x = 0.1$ they are slightly optimistic, by roughly 5–10%, which shows up as coverage a few points below ideal. So with synthetic data we have confirmed the tool we will rely on for real data, where only one sample is available.

8.4 The detection limit

Finally, we repeated the experiment with detection limits $r_{\text{det}} = 1$, 2 and 3 μm, dropping all bins below the limit. For $D_x = 0.1$ we used $V = 1$ cm^3, so that the no-cutoff error is small enough to see the effect clearly.

	$D_x = 10$, $V = 0.1$ cm^3			$D_x = 0.1$, $V = 1$ cm^3		
cutoff	droplets seen	predicted	observed	droplets seen	predicted	observed
none	2053	0.061	0.064	34387	0.071	0.079
1 μm	1977	0.063	0.060	31437	0.122	0.119
2 μm	1870	0.066	0.069	28072	0.247	0.266
3 μm	1705	0.071	0.070	23879	0.352	0.561

Table 4: Spread of $\hat{\ell}$ (approximately relative error) with a detection limit, 200 samples each.

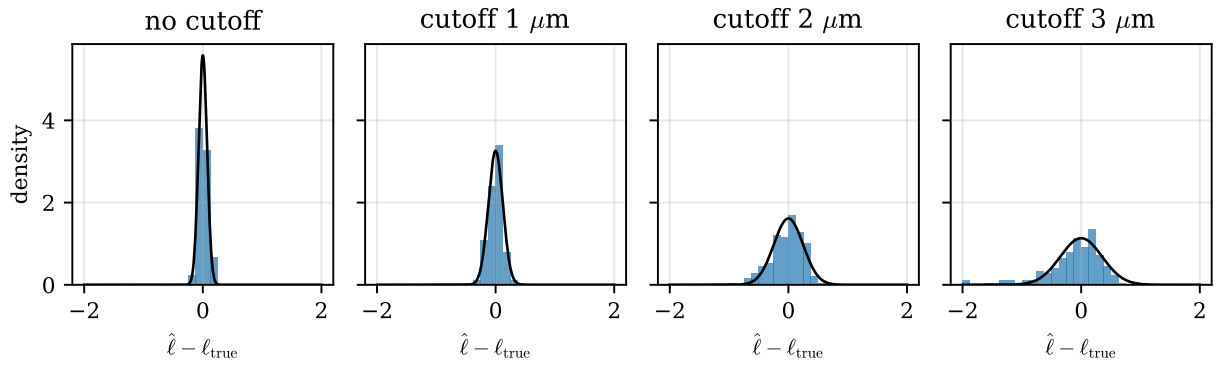


Figure 8: Distribution of $\hat{\ell} - \ell_{\text{true}}$ for $D_x = 0.1$, $V = 1$ cm^3, as the detection limit rises, with the predicted Gaussian.

- **For $D_x = 10$ the cutoff barely matters.** Strong broadening reshapes the whole distribution, so seeing most of it is enough. Even a $3\,\mu$m cutoff, which hides 17% of the droplets, raises the error only from 6.1% to 7.1%.
- **For $D_x = 0.1$ the cutoff is severe.** A $2\,\mu$m cutoff hides only 18% of the droplets but more than triples the error, from 7% to about 25%. With a $3\,\mu$m cutoff the observed spread is about eight times the no-cutoff value, and the estimates are skewed and the curvature error bar is optimistic: predicted 35%, observed 56%, the same low-information breakdown seen in Section 8.2. When turbulence is weak, its fingerprint is mostly fine structure near the activation size of about $1\,\mu$m, which is exactly what the cutoff hides.

So whether D_x can be recovered from instrument data depends strongly on where the detection limit sits relative to the activation region, and on how large D_x is.

9 Summary

- Sampling a small volume of a well-mixed chamber gives unbiased but noisy counts. Each bin's count is Poisson with mean $\mu_k = V \int_{\text{bin}} N \, \mathrm{d}x$, and different bins are independent.
- Maximum likelihood gives the Poisson deviance (5) as the loss. Unobserved bins (below the detection limit) are left out, not set to zero.
- The adjoint machinery is unchanged except for $\mathbf{g} = V\mathsf{H}^\top[2(1 - c_k/\mu_k)]$.
- The curvature of the loss at its minimum gives a single-sample error bar, $\mathrm{sd}(\hat{\ell}) \approx \sqrt{2/\mathcal{J}''}$. Over many synthetic samples, these error bars match the observed scatter and contain the truth about 68% of the time, as they should.
- Precision improves as $1/\sqrt{V}$. Small D_x is intrinsically harder to pin down, and a detection limit above the activation size makes it much harder. When the information is low, the estimates become skewed and the error bars become optimistic.

Natural next steps are to generate the truth on a finer grid than the fit (so that numerical errors no longer cancel), to add noise that is correlated across bins or larger than Poisson, and to estimate additional parameters alongside D_x.

A Parameter values

Quantity	Value	Note
Temperature, pressure	288 K, 1000 hPa	
Supersaturation	$s = 0.09\%$	
Growth coefficient	$a = 205.8\,\mu\mathrm{m}^2/\mathrm{s}$	$\mathrm{d}(r^2)/\mathrm{d}t = a\,s$ for large droplets
Köhler parameters	$A_k = 1.128 \times 10^{-3}\,\mu\mathrm{m}$, $B_k = 3.650 \times 10^{-4}\,\mu\mathrm{m}^3$	NaCl, dry radius $0.065\,\mu\mathrm{m}$
Removal constant	$k = 1.210 \times 10^{-4}\ \mu\mathrm{m}^{-2}\mathrm{s}^{-1}$	Stokes fall speed / chamber height (1 m)
Injection rate	$F_{\mathrm{in}} = 100\ \mathrm{cm}^{-3}\mathrm{s}^{-1}$	
Grid	8000 cells, $\Delta x = 0.1125\,\mu\mathrm{m}^2$, $r \in [0.0325, 30]\,\mu\mathrm{m}$	
Bins	8 cells per bin, width $0.9\,\mu\mathrm{m}^2$, $K = 1000$	
Sample volumes	$V = 0.01$, 0.1, 1 cm^3	
Monte Carlo	200 samples per case	starting guess $D_x = 1$, bounds $[10^{-3}, 10^3]$